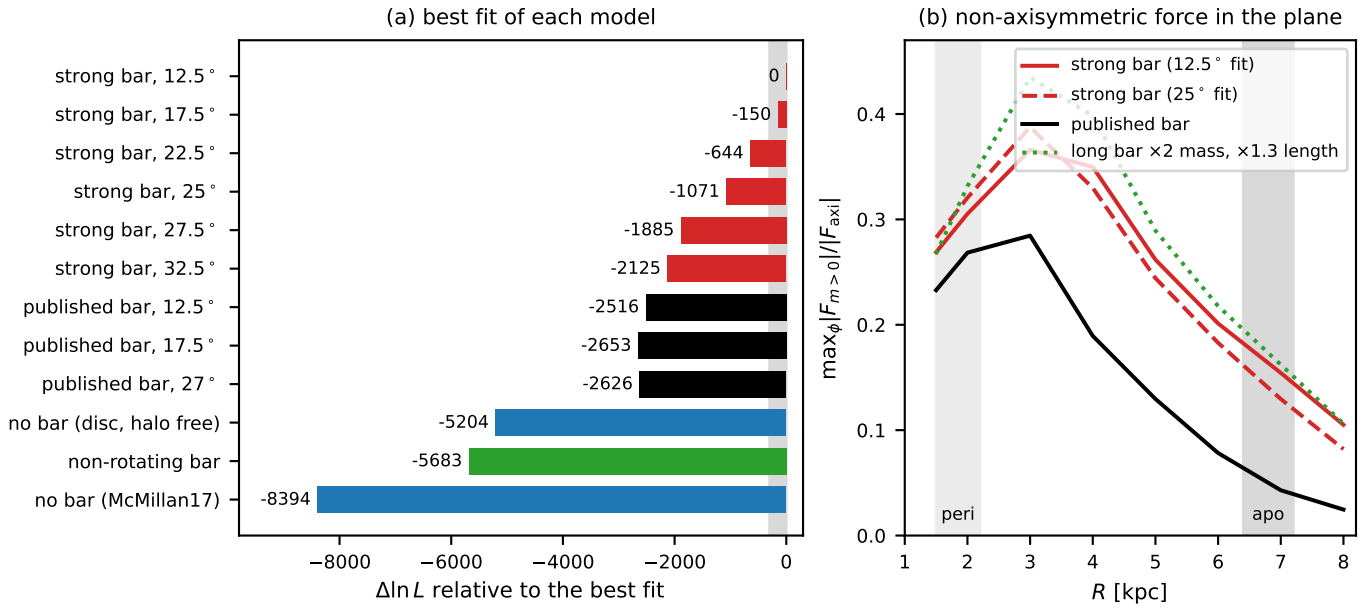
(a) best fit of each model
strong bar, 12.5°
strong bar, 17.5°
strong bar, 22.5°
strong bar, 25°
strong bar, 27.5°
strong bar, 32.5°
published bar, 12.5°
published bar, 17.5°
published bar, 27°
no bar (disc, halo free)
non-rotating bar
no bar (McMillan17)
0
-150
-644
-1071
-1885
-2125
-2516
-2653
-2626
-5204
-5683
-8394
−8000
−6000
−4000
−2000
0
ΔlnL relative to the best fit
(b) non-axisymmetric force in the plane
strong bar (12.5° fit)
strong bar (25° fit)
published bar
long bar ×2 mass, ×1.3 length
max$_\phi$|$F_{m>0}$|/|$F_{\rm axi}$|
0.0
0.1
0.2
0.3
0.4
peri
apo
1
2
3
4
5
6
7
8
R [kpc]